

Question number	Answer	Additional guidance	Mark
(a)	An explanation that includes four of the following points: • with (increasing) age the number of dopamine producing cells decreases (1) • after {44/45} years the number of dopamine producing cells fall below {normal / people without Parkinson's / 350 000} (1) • {less / fewer} {neurotransmitters / dopamine} {released / produced} (1) • therefore {fewer / no} {(electrical) impulses / action potential / nerve impulses} (1) • causing {reduced / no / disruption to} {muscle contraction / responses} (1)	ACCEPT negative correlation ACCEPT identification of age where number of dopamine producing cells is below {normal / people without Parkinson's / 350 000} ignore {no / lack of} {dopamine / neurotransmitter} ignore L-Dopa is hard to convert into dopamine ACCEPT less depolarisation ACCEPT threshold value not met ACCEPT causes paralysis ACCEPT loss of control of muscles / loss of motor {control / function} ignore increase in symptoms	(4)

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(b)	An answer that includes two of the following points (carbidopa) prevents L-DOPA being broken down (in digestive system / before blood-brain barrier) (1) (because) L-Dopa can pass blood-brain barrier / dopamine can't pass blood-brain barrier (1) - allowing a lower dose of L-DOPA to be given (1)	ACCEPT less L-DOPA {converted / broken down} to dopamine (in digestive system / before blood-brain barrier) ACCEPT (more) L-DOPA {converted / broken down} to dopamine {in the brain / after the blood-brain barrier} ACCEPT more L-DOPA {absorbed into blood / transported by blood (to brain)} ignore dopamine should only be in brain and not digestive system ignore enter brain unqualified L-Dopa shouldn't be converted to dopamine until after it has crossed the blood brain barrier = 2 marks	(2)

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(c)	An answer that includes the following points: - as dopamine concentration increases the (mean) heart rate increases and then decreases (1) - correct comment on mean heart rate increasing up to { 3 / 4 } / decreasing after 4 or at 5 } (1) - correct comment on error bars and {significant difference / no significant difference / significant effect} (1) - accept relevant comment on methodology (1)	ACCEPT mean heart rate doesn't change between 0-1 e.g. between log {0-1 / 0-2 / 1-2 / 3-4} error bars overlap therefore there is no significant {difference / effect} e.g. between log {2 / 3} error bars do not overlap therefore there is a significant difference e.g. error bars at 5 don't overlap with any others so results are significantly different ignore error bars overlapping without being specific as to which e.g. only 10 flies used/ no indication of age / sex / environmental conditions / only tested on one species / small sample size	(4)

